

Preliminary Applications of Machine Learning to Circumbinary Planets - A New Approach for Recalcitrant Transiting Worlds

REMY LIU

ABSTRACT

Circumbinary systems are one of the most exotic exoplanetary architectures discovered thus far, in which planets orbit not one star but the centre of mass between two mutually orbiting stars at great distance. Between 10 and 15 circumbinary systems are currently known, depending on the status of a few candidates, far less than the 5000+ exoplanets confirmed in single-star systems. Nevertheless, they present a exotic and potentially significant environment for planetary formation and evolution and are thought to be significantly different from single-star systems at the population level, with fewer massive planets and an excess of highly-inclined orbits.

Transiting circumbinary systems, ones where the planet eclipses one or both stars at certain parts of its orbit, make up the majority of all known circumbinary systems. Transiting planets around a single star can often be detected through fully automated means since their orbits are exactly periodic or very nearly so. However, the asymmetric gravitational potential of a binary host causes planetary orbits to shift rapidly, with period variations on the order of 10-20 days across a 100-day orbit. Worse still, the rapid precession of the orbit's inclination can shift a planet out of the transit window intermittently, causing transits to be irregularly 'skipped'. Orbits very close to the central binary are also unstable, so circumbinary planets tend to have orbital periods on the order of hundreds of days, resulting in a small total number of transits over any given observation period. Thus, detection of these systems thus far has almost entirely relied on manual searches, picking out large circumbinary planets by visual inspection. Such searches are however exceptionally time-consuming and likely to miss smaller planets.

Using a convolutional neural network (CNN), I have conducted a survey of all 2920 eclipsing binaries observed by the transit-finding telescope Kepler. This is the largest survey dedicated to searching for circumbinary planets of any kind thus far and the only one to use a CNN-based technique. I was able to reduce the visual search volume by an excess of 90% compared to a solely manual approach and detected five novel candidate transiting circumbinary systems, a significant proportion relative to the total number detected thus far.

I am able to efficiently investigate eclipsing binaries for transiting circumbinary planets due to my CNN-based technique's complete non-reliance on periodicity, allowing me to find irregular transit events that more traditional automated techniques struggle to detect. Similar approaches could be used to efficiently search large numbers of systems for other non-periodic events of interest, such as stellar flares or exo-cometary transits.

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1. INTRODUCTION

1.1. *Background*

Conjectures about the existence of planetary systems around other stars have been made in philosophical and scientific circles for about as long as the analogy between our Sun and the stars have been made (Maor 2013). In the past 30 years, such conjectures have been soundly vindicated with the discovery of several thousand extrasolar planets, ranging from slightly larger than the Moon (Konacki and Wolszczan 2003) to several times the mass of Jupiter (Nelson and Angel 1998). Exoplanetary systems may include several planet types not observed in the Solar System, including very short-period gas giants (Mayor and Queloz 1995), planets intermediate in mass between our terrestrial planets and ice giants (Rivera et al. 1995), and planets with exceptionally eccentric orbits resembling those of comets (Fossey et al. 2009), among others. Of particular interest to both specialists and the public are habitable-zone rocky planets similar to Earth, which may provide life-supporting conditions (Borucki et al. 2012). The great diversity of exoplanets observed suggests there exists a diverse yet poorly understood set of formation mechanisms that contribute to the exoplanet population we observe (Manara et al. 2018)(Lin et al. 2021), mechanisms that can evidently produce significant populations of planets even in seemingly hostile environments (Dupuy et al. 2016). Investigation of these types of systems may reveal further insights about the modes of planetary formation abundant in the universe, which may aid precision searches for targets of particular interest (e.g. Earth-like rocky planets).

Circumbinary exoplanet systems are some of the most exotic of those observed thus far. Though commonly depicted in fiction, they were assumed to be impossible in reality for a substantial period due to the belief that the nonuniform gravitational potential of the host would make planetary orbits chaotic to the point of instability (Harrington 1981).

However, improved hydrodynamic simulations (Childs and Martin 2021) and detections of circumbinary planets and protoplanetary disks (Elsender et al. 2021) suggest they should be fairly common. The circumbinary exoplanet population is expected to be significantly physically different than the general exoplanet population; a physical absence of massive Jupiter-size objects is identified even with the low number of known systems (Welsh et al. 2014). Investigation of this exotic and likely hostile environment to planetary formation is likely informative of the robusticity of planet formation as a whole, while habitable, Earth-like worlds may be found in circumbinary systems as planets or moons of large gas giants (Popp and Eggl 2021).

1.2. Past Work

Previous discoveries of circumbinary systems have occurred through a variety of means. One planet each has been found through pulsar timing (Thorsett et al. 2014) and radial velocity (Standing et al. 2023), and a small number of directly-imaged objects that probably did not form through disk accretion like typical planets (Rodet et al. 2017)(Currie et al. 2014)(Janson et al. 2021). A fair number of planet candidates have been identified orbiting eclipsing post-common envelope binaries through the eclipse timing method (Pulley et al. 2022), but the majority of these have been disputed (Krzesinski et al. 2020) or disproved (Pulley et al. 2022), suggesting that the eclipse timing variations used to justify their existence was caused by other mechanisms. However, the majority of known circumbinary planets have been discovered through the transit method, which detects planets using dips in stellar light output caused by its obscuration by a planet’s disk, which does not emit significant visible light (Charbonneau et al. 1999). The existing sample of transiting circumbinary systems mainly consists of Kepler Space Telescope objects, as does the population of known transiting exoplanets as a whole.

Kepler was a 1.4-meter space telescope in operation between the years 2009 and 2018, designed to continuously monitor the flux of a patch of around 100,000 stars in the constellations Cygnus and Lyra (Siddiqi 2018) in an effort to detect transiting exoplanets between 0.5 and 2 Earth radii within the habitable zones of Sun-like stars, which were largely undetectable by methods such as radial velocity (Meunier and Lagrange 2013). Kepler took continuous 90-day exposures of its survey field at 30-minute cadence for a primary mission time of just over 4 years, producing a continuous record of stellar flux for approximately 13.2 million targets (“KIC Search Help”). This data is publicly available through the Mikulski Archive for Space Telescopes (MAST), which also compiles a catalogue of Kepler eclipsing binaries which I used to locate the data for all systems of interest targeted by my survey.

Out of the 5000 eclipsing binaries in the Kepler primary mission’s visual field, 12 confirmed planets in 10 systems are known. These low numbers are probably not due to actual rarity (Elsender et al. 2021) but rather because of low survey completeness (Welsh et al. 2013) and technical challenges associated with the nature of circumbinary planets. Thus far, all surveys dedicated to the detection and confirmation of circumbinary planets have been limited to under 1000 targets (Welsh et al. 2013) and detection has been achieved mainly by human visual examination.

2. RESEARCH OVERVIEW

In this research, I strive to create a fully-automated detection algorithm that is able to locate circumbinary planets and other non-periodic features in Kepler data. To accomplish this, I take a machine learning-based approach which encompasses the following:

1. I consult the literature for the properties of known circumbinary exoplanets, including mean and variability of transit durations, depths, SNR, and transit-timing variations.
2. I generate a large artificial dataset which resembles real transits based on this information, which is used to train a Convolutional Neural Network to identify singular transit features.
3. I apply the trained Convolutional Neural Network to the full Kepler eclipsing binary catalogue to identify systems in which possible transit features are observed.
4. I assess the validity of candidate systems recovered by the Convolutional Neural Network to determine the efficacy of my method compared to traditional visual recovery.

3. CONVOLUTIONAL NEURAL NETWORK

3.1. Why Neural Networks?

Circumbinary planets present several challenges to detection not shared by single-host systems. Efficient automated surveys for them are inhibited by a number of factors, primarily transit-timing variations (TTVs). The nonuniform gravitational potential of the host binary results in significant perturbations to the planetary orbit and large, non-Keplerian fluctuations in several orbital parameters including the period, inclination, and apsidal locations. While this can help distinguish a planetary signal from background sources such as unresolved additional eclipsing binaries (Bienias et al. 2021), it also prevents automated period-based investigations from finding additional transits, precluding addition to automatically compiled planet candidate sets (Welsh et al. 2013). Further confounding the issue is the fast precession of circumbinary planets’ orbital parameters, oftentimes resulting in intermittent shifting in an out-of-transit window that causes expected transits to be skipped, sometimes for long periods (Kostov et al. 2014). As such, nearly all searches for circumbinary planets thus far have relied on manual detection (Welsh et al. 2013)(Kostov et al. 2020)(Doyle et al. 2011)(Schwamb et al. 2013)(Welsh et al. 2015)(Kostov et al. 2016)(Orosz et al. 2012a)(Orosz et al. 2012b), with only one attempt at automated transit detection (Kostov et al. 2014) made thus far. Manual searching is inherently biased towards large planets, which produce transits comparable in size to stellar eclipses and thus stand a chance of being noticed in the light curve. Additionally, large sample sizes make manual inspection prohibitively time-consuming unless the search is outsourced as was the case for the quadruple system Kepler-64 (Schwamb et al. 2013).

A Convolutional Neural Network (CNN) offers significant advantages over a manual search. CNNs can achieve high detection accuracy at high speed, even within complex situations such as image recognition (Russakovsky et al. 2015). The Kepler light curve data, as a 1-D flux array, is a broadly similar data format in which these advantages are expected to exist as well. The difficulty of manually examining all candidates is significantly reduced by the elimination of the vast majority of nondetections by the CNN and it is not subject to bias toward larger candidates so long as they are distinguishable from noise. Furthermore, there is no periodicity requirement in this method, so the issues of TTVs and precession-driven migration in and out of the transit window are completely nullified. Furthermore, long-period transiting objects which cannot be confirmed from a small number of transits may also be identified by this method.

3.2. *Network Specifics*

I used a relatively standard keras CNN as a base and added a pair of circular convolution layers. The purpose of these is to provide wraparound capability, allowing for at least some recovery in the case where a transit is cut off by the end of the light curve array; wraparound allows for the other side of the array to complete the transit.

My CNN takes a 240-element flux array and outputs a single number from 0 to 1 which is the probability that the flux array contains a transit feature. To increase transit visibility and model ability, all inputs are scaled by a factor of 2σ , where σ is the measured noise. This allows me to make use of the model’s whole parameter space, improving overall recovery as compared to the raw light curve, while also maintaining similar relative scale for all transits. Altogether, there are 329,217 tunable parameters (See Table 1).

3.3. *Training*

Assembling training samples from transits of known exoplanets, circumbinary or otherwise, may be possible, but it is time-consuming to compile the requisite large number of sample transits. Moreover, training with the existing sample of circumbinary transits runs the risk of overfitting, where the CNN learns to distinguish only the contents of the training set and provides no useful predictions with novel data.

A simulated transit is generated with a simple trapezoidal approximation by passing the desired transit duration, depth, and the size ratio between planet and star. A 240-point noise array is generated at a randomly determined SNR using Gaussian sampling. This noise array is used unmodified as a negative sample, while it is added to the simulated transit to form a positive sample, forming a pair.

Visual examination is likely to reveal Jupiter-size circumbinary planets (Welsh et al. 2013)(Kostov et al. 2016) so the model was trained with simulated SNRs between 2 and 8 to focus the search primarily on small transits not so easily distinguishable from noise. Noise levels vary significantly between systems and within a single light curve due to various factors such as binary-driven stellar activity (Johnstone et al. 2019) or varying Kepler noise, so a large range of noise levels are generated. Among known circumbinary systems, transit duration is on the order of 0.1 to 0.5 days (37) while the size ratio between detectable planets and Sun-like stars is on the order of 0.001 to 0.1. Transit depth

Table 1. CNN Detailing

Layer Name	Number of Parameters	Output Shape
Input	0	[(None, 240, 1)]
Gaussian Noise	0	(None, 240, 1)
Reshape	0	(None, 1, 240, 1)
Cropping 2D	0	(None, 1, 1, 1)
Cropping 2D	0	(None, 1, 1, 1)
Concatenate	0	(None, 1, 242, 1)
2D Convolution	512	(None, 1, 240, 128)
Cropping 2D	0	(None, 1, 2, 128)
Cropping 2D	0	(None, 1, 3, 128)
Concatenate	0	(None, 1, 245, 128)
2D Convolution	196864	(None, 1, 240, 256)
Reshape	0	(None, 240, 256)
1D Global Max Pooling	0	(None, 256)
Flatten	0	(None, 256)
Dense	65792	(None, 256)
Dense	65792	(None, 256)
Dense	257	(None, 1)

NOTE—A more specific detailing of my CNN. In total, there are 329,217 parameters in this model.

is left in its default range of 0.0001 to 0.01 since the $2\text{-}\sigma$ standard scaling renders all inputs to comparable size (See Figure 1).

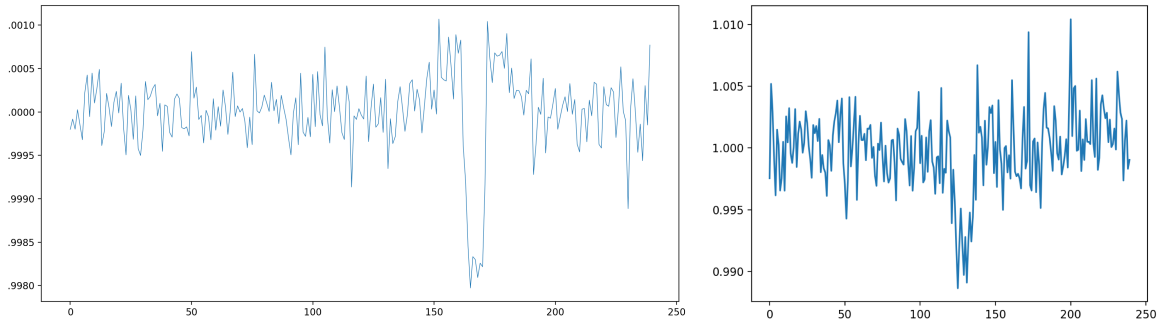


Figure 1. A randomly selected artificial training data sample, compared to a transit of the confirmed planet Kepler-35 b [KIC 9837578]. Despite differing depths, SNR and appearance are similar. The latter transit was successfully recovered by the CNN, among others.

An initial sample of 5,000 positive-negative pairs was generated as a training sample, while an additional 1,000 were generated as a preliminary verification dataset. This is relatively low for average CNN operations but the complexity of the situation is relatively low and initial experimentation suggested larger training samples would be excessive.

To train the CNN, I input 32 training samples at a time into the model and used the Adam optimiser to optimise the model (Kingma and Ba 2017) for 80 epochs. The smaller batch size helps prevent overfitting, while the relatively large number of epochs is used to ensure consistent high accuracy. Training proceeded without incident, though 80 epochs proved to be excessive (See Figure 2).

3.4. Results

Within the 1,000-item verification sample, no false positives were observed. To rectify this and to obtain a better estimate of true rates of inaccuracy, another set of 100,000 synthetic light curves were generated using the same generation algorithm and evaluated with the model (See Figure 3). This test reveals exceptionally low false positive

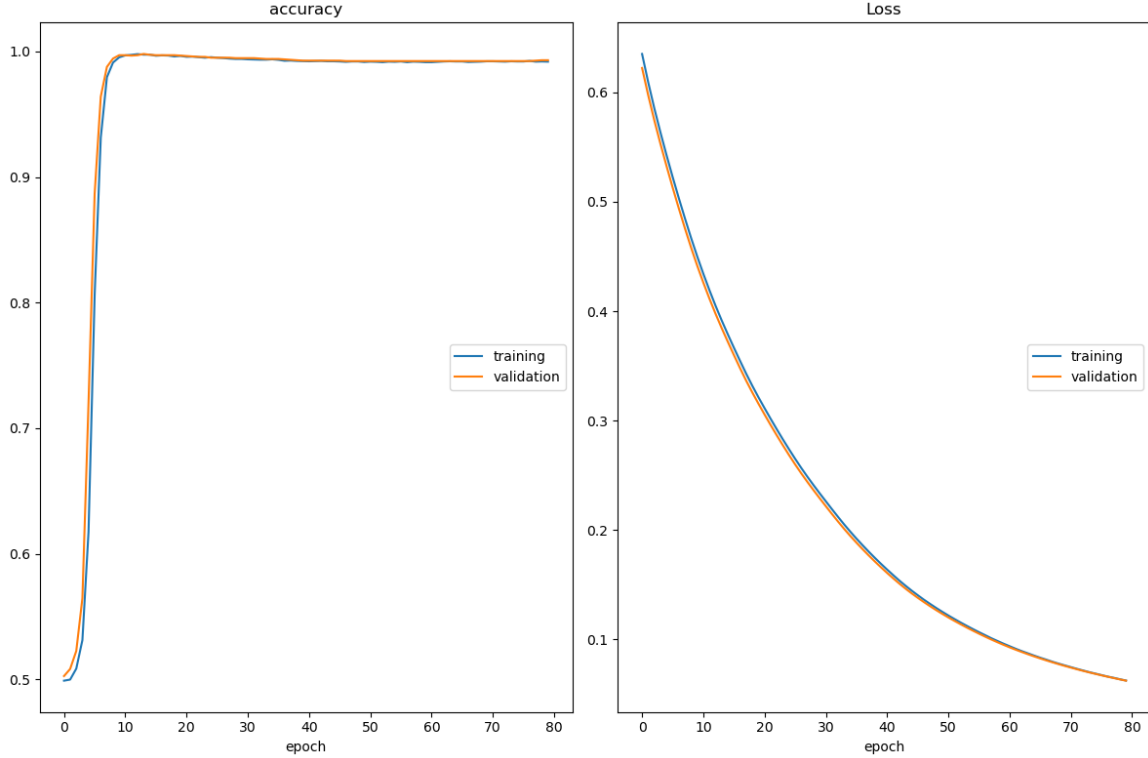


Figure 2. Training curve for the model. 80 epochs was ultimately excessive and declines due to over-training occurred. The final model parameters used were from the most accurate epoch observed.

Table 2. Model Performance

Parameter	Value
Validation Accuracy	0.99971
True Positive Rate	0.99973
False Positive Rate	0.00031

NOTE—Numerical results of the CNN performance on the 100,000-sample simulated dataset.

and false negative rates (See Table 2), allowing large data volumes to be processed without problematic excesses of false positive detections or loss of true candidates.

4. FULL PIPELINE

4.1. Preprocessing

The 2920 eclipsing binaries in the Kepler field are listed in the MAST Kepler Eclipsing Binaries catalogue ("Kepler Eclipsing Binaries"), which was used as a basis for my investigation.

Of the known transiting circumbinary systems, the three-planet system, Kepler-47, has the shortest stellar orbit at 7.45 days (Orosz et al. 2012b). There is no physical obstacle to the observation of circumbinary transits in systems with lower periods (Fleming et al. 2018) but no planets are known nonetheless, suggesting there are physical limitations

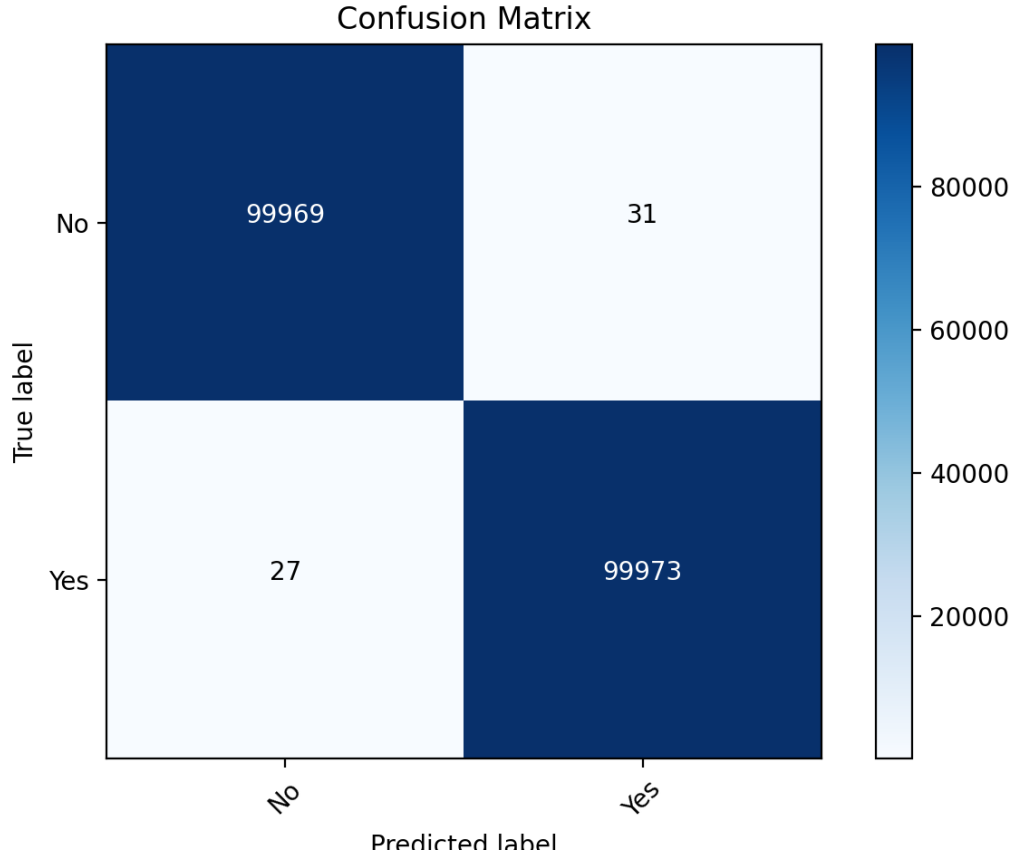


Figure 3. Numerical confusion matrix of the full 100,000-sample simulated verification dataset.

on the presence of planets in such systems. Indeed, simulations find that short-period binaries tend to eject close-in circumbinary planets as the stability limit migrates during planet formation (Fleming et al. 2018). There are only two extant reports of circumbinary planets in binaries with periods below Kepler-47’s, that being the post-common envelope cataclysmic binary DP Leonis (Beuermann et al. 2011) and the low-mass X-ray binary MXB 1658–298 (Jain et al. 2017), but neither of these highly-evolved systems resembles typical circumbinary planet hosts, nor are the planets’ periods short enough to permit use of the transit method even if they were transiting (Beuermann et al. 2011)(Jain et al. 2017). Within the Kepler Eclipsing Binaries Catalogue 1777 systems have a period of less than 5 days, so they were excluded from the initial search. This left 1143 systems to be investigated.

For all systems not excluded by the 5-day limit, light curve data was downloaded from the Kepler primary mission data release. This data was detrended to remove stellar activity, as is standard. The detrending algorithm uses a smoothing spline, attempting to minimise the Bayesian information criterion by adjusting its smoothing rate. Prior to detrending, I mask all known transits (See Figure 4), which the stellar eclipses generally would have been flagged in as a later-determined false positive. Outliers detected during detrending are removed to avoid cosmic ray signals, excessive Kepler noise, and other such interference.

4.2. Neural Network Usage

Additional data points with a value of 1 were appended to the ends of all light curves until evenly divisible by 240, upon which they were split into 240-point segments. All segments were transformed as stated in 3.2-I for input into the CNN. If the model identified a segment with greater than 0.9 confidence, the segment was recorded and the system was flagged as a transit candidate.

An additional transit-flagging method was used on all light curves as well to test its utility for identifying smaller features. This method consisted of flagging all portions of the light curve array that contained at least 3 consecutive points (90 minutes of data) $3\text{-}\sigma$ below the mean.

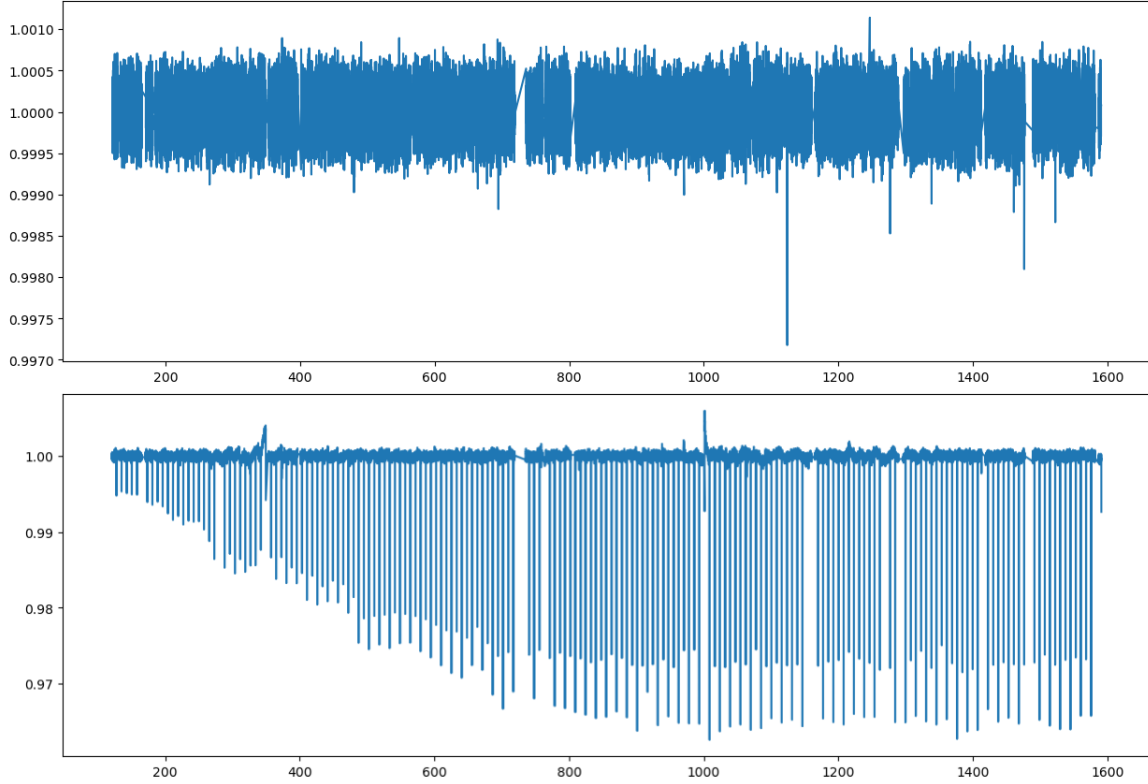


Figure 4. Pre- and post-masking and detrending of a single sample system (KIC 7955301). This algorithm is applied to all systems within the restricted catalogue.

4.3. Post-Processing

Following the CNN selection, 579 systems were flagged as novel candidates. All candidates were examined visually to investigate their validity. Transit candidates flagged by the CNN were classified as either ‘Noise’, referring to Kepler red noise that was incorrectly flagged, ‘Contact Binary’ for semidetached and overcontact systems whose light curves had been erroneously detrended, ‘Uncertain’ for small or very short-duration features whose identity was unclear, and ‘Likely Transit’ (See Figure 5) for features of sufficient duration and correct shape to be identified as transits with some confidence. Systems were sorted into categories based on the number of questionable or likely transits identified in their light curves and the identifications of all transit candidates in non-erroneously processed systems were recorded.

5. RESULTS

5.1. Novel Candidates

A total of 113 KICs were marked as follow-up targets on the grounds of containing at least one CNN-identified segment manually flagged as a ‘Likely Transit’. Of these, 83 were background binaries as shown by the exact periodicity of the transit events upon manual investigation. Of the remaining 30, a further 21 were produced by masking errors and also excluded, leaving 10 objects of relative confidence. The majority are known from only one or two candidate transits so I refrain from identifying them as planets, but their sizes suggest they are planets.

For objects less than 0.1 times the size of the eclipsed star, it is permissible to ignore limb darkening at the base of the transit, since the area of the stellar surface covered is small enough that it has approximately consistent luminosity (Mandel and Agol 2002). Furthermore, the ratio of stellar radius to semimajor axis is necessarily large for circumbinary planets, which are unstable at small orbital radii (Welsh et al. 2014), so the depth of a transit is controlled essentially only by the impact parameter. Estimating impact parameter for individual transits may be possible with sufficient computational time, but was judged as too time-costly for a preliminary examination such as this one. As such, the radius calculations assume the impact parameter is zero, providing a lower bound on the transiting objects’ sizes. Stellar magnitudes and temperatures were obtained through Gaia DR3 (“Gaia DR3”), from which stellar radius was

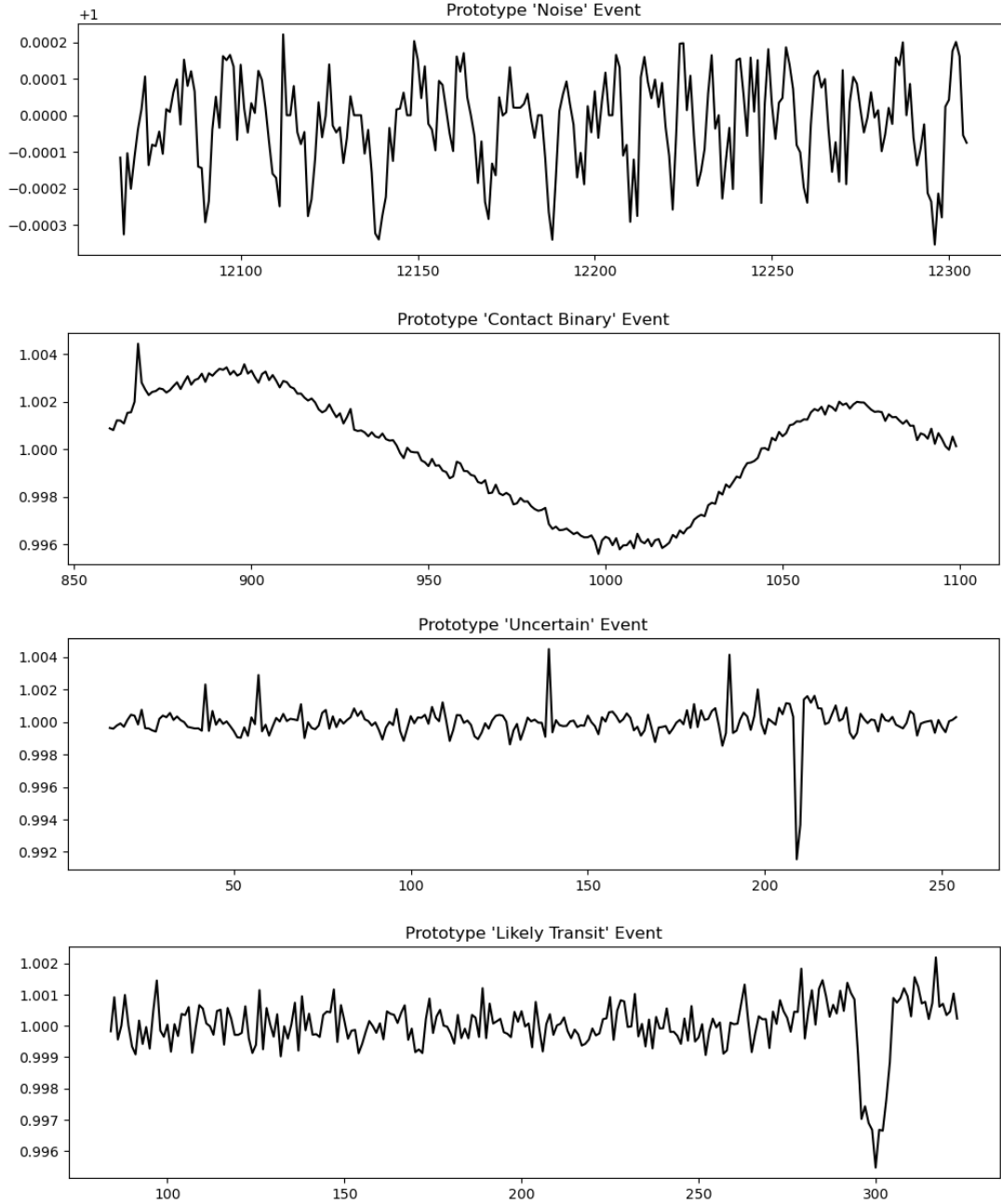


Figure 5. A ‘Noise’, ‘Contact Binary’, ‘Likely Transit’, and ‘Uncertain’ event, in that order. The ‘Noise’ event is from a candidate-less system, KIC 4247991, while the ‘Contact Binary’ signal is an erroneously processed semi-detached system, KIC 8618226. The ‘Likely Transit’ is noticeably wide and has a flat bottom (noise notwithstanding), while the ‘Uncertain’ event is narrow despite being much larger than surrounding noise. Both these latter events come from the confirmed circumbinary system KIC 9837578 (Kepler-35).

estimated through the Stefan-Boltzmann Law. The proportion of stellar flux obscured is thus directly proportional to the cross-sectional area ratio of the planet and star, so the planetary radius can be calculated (See Table 3). All candidates recovered, should they exist, have radii significantly less than that of Jupiter and thus must be of planetary mass.

KIC 5983532 (KOI-4289) is a rotating ellipsoidal variable. Its MAST period is given as 147.288448 ± 0.001373 days but this is almost certainly erroneous based on its identity as an ellipsoidal variable. A single candidate transit is recovered by the CNN at 1430 days (See Figure 6).

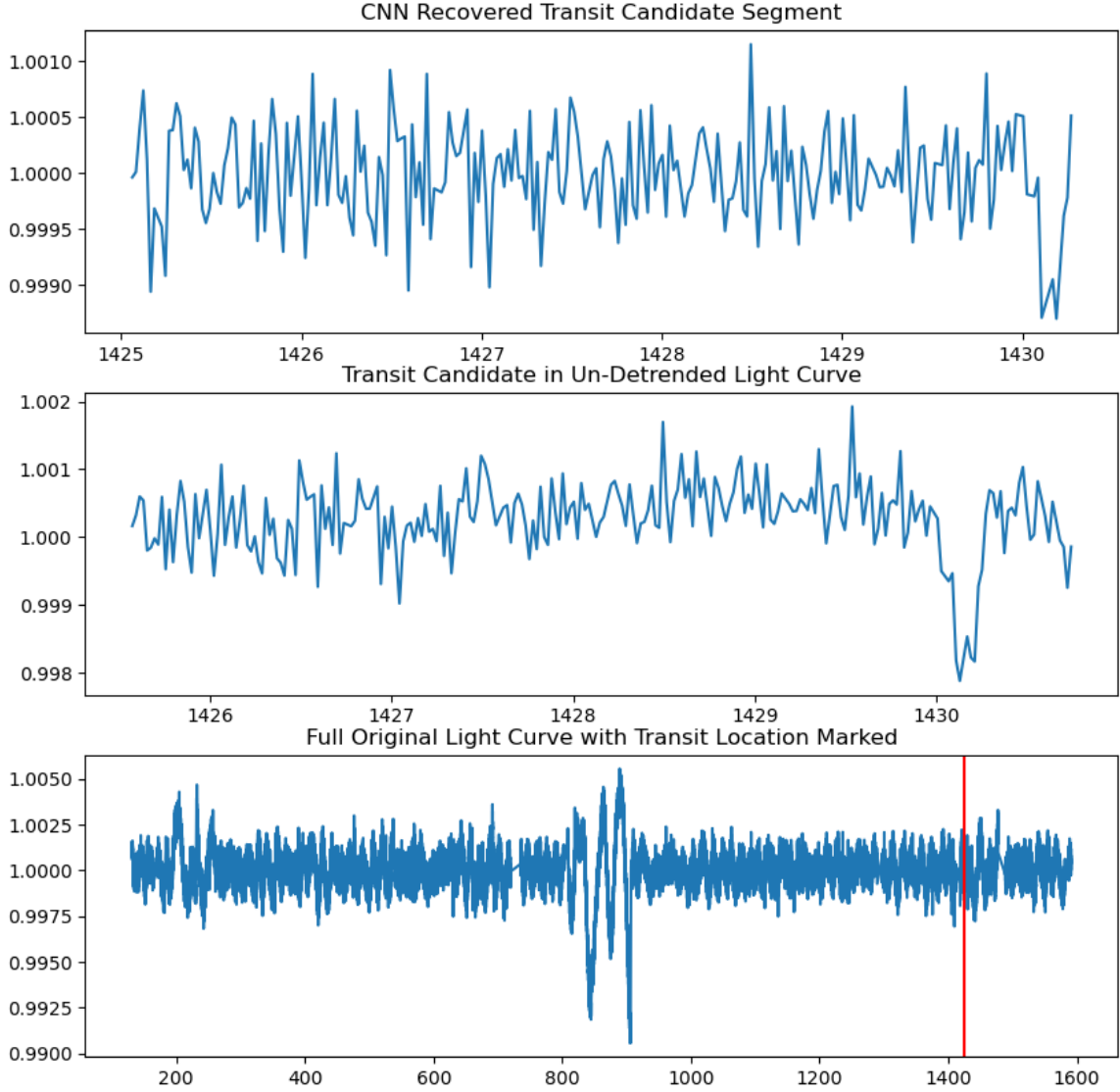


Figure 6. Figure 6: The light curve segment containing the identified transit candidate in KIC 5983532. Both the detrended light curve used to identify the candidate and the un-detrended original. The position of the candidate within the full original light curve is also marked. The un-detrended window is shifted right by 0.5 days in order to avoid clipping the end of the transit candidate.

KIC 6672229 (KOI-6754) is an eclipsing binary with a MAST period of 23.832374 ± 0.000108 days. A single transit candidate unassociated with the stellar eclipses is reported by the CNN at 634 days (See Figure 7).

KIC 7136958 (KOI-3350) is an eclipsing binary with a MAST period of 29.015853 ± 0.00017 days. A single transit candidate partially cut off by a data gap is observed at 537.6 days (See Figure 8).

KIC 7668648 (KOI-6899) is an eclipsing binary with a MAST period of 27.81859 ± 0.000126 days. Three CNN-identified transit candidates were reported at 946.5, 1276.5, and 1345.2 days (See Figure 9). This system is notable for the continually deepening stellar eclipses and also contains a third star that mutually eclipses the other two at a period of 204.8 days (Zhang et al. 2017). It is unknown whether the three events are from a single planet without four-body modelling, which may not be possible due to the uncertain orbit of the tertiary star. However, known

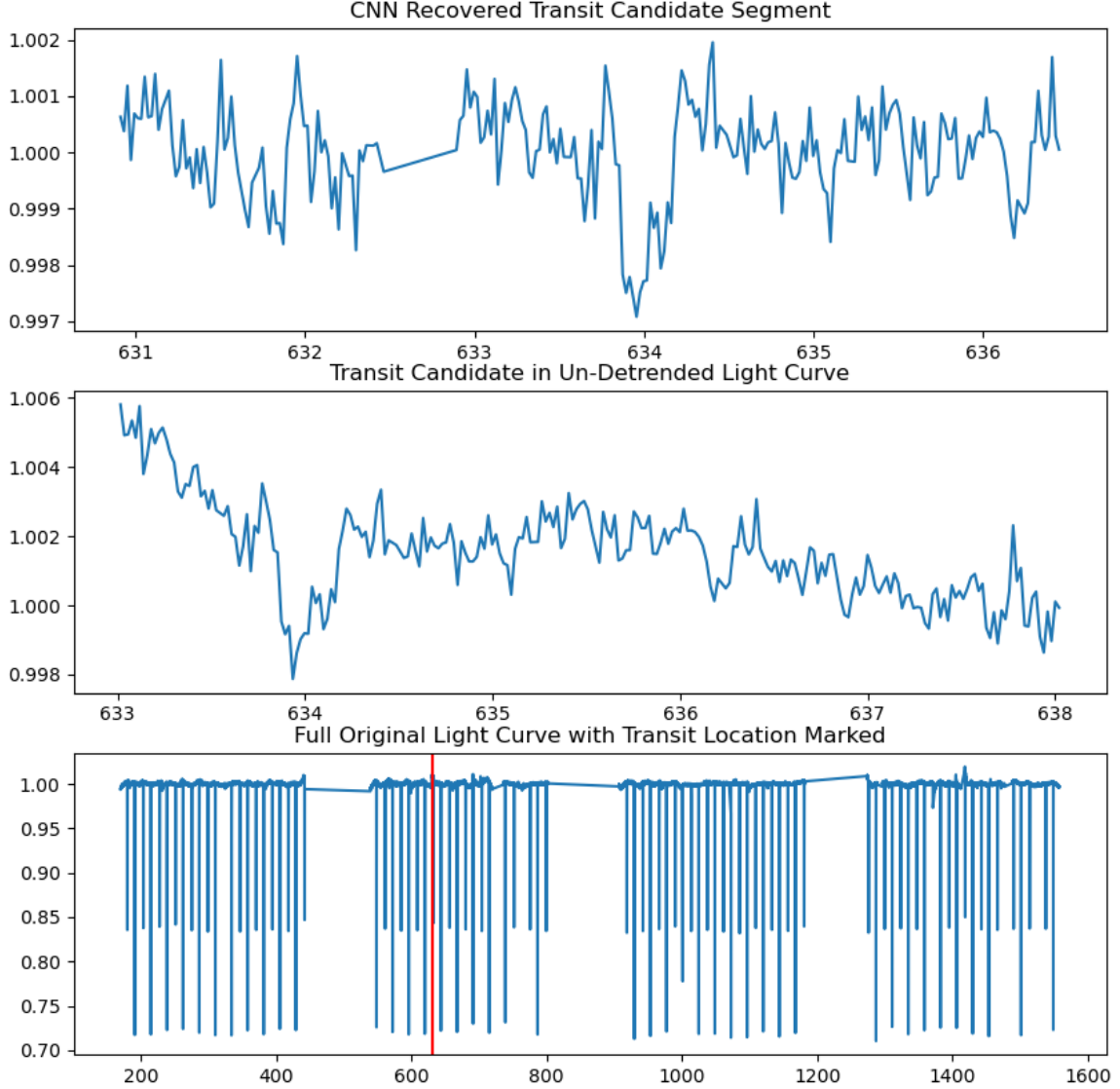


Figure 7. The light curve segment containing the identified transit candidate in KIC 6672229. Both the detrended light curve used to identify the candidate and the un-detrended original. The position of the candidate within the full original light curve is also marked. The un-detrended window is shifted right by 2 days in order to exclude a primary eclipse masked in the detrended sample.

stability limits for P-type and S-type systems would suggest that none of the transits can belong to the same planet due to the resultant orbits falling in the instability zone of either the central binary or the tertiary.

KIC 11619964 (KOI-7466) is an eclipsing binary with a MAST period of 10.368555 ± 0.000031 days. It contains a single transit candidate identified by the CNN at 1400.8 days, though this occurs in a region of high noise and thus the identification may be questionable compared to other candidates reported.

5.2. Sources of False Positives

Though the majority of signal-less systems were excluded from the dataset, there were still a significant number of erroneous detections.

In some systems, the noise level is generally excessive or there are patches of excessive noise, resulting in patterns that vaguely resemble transits (See Figure 6). The CNN is phase-insensitive and may be confusing such patterns with

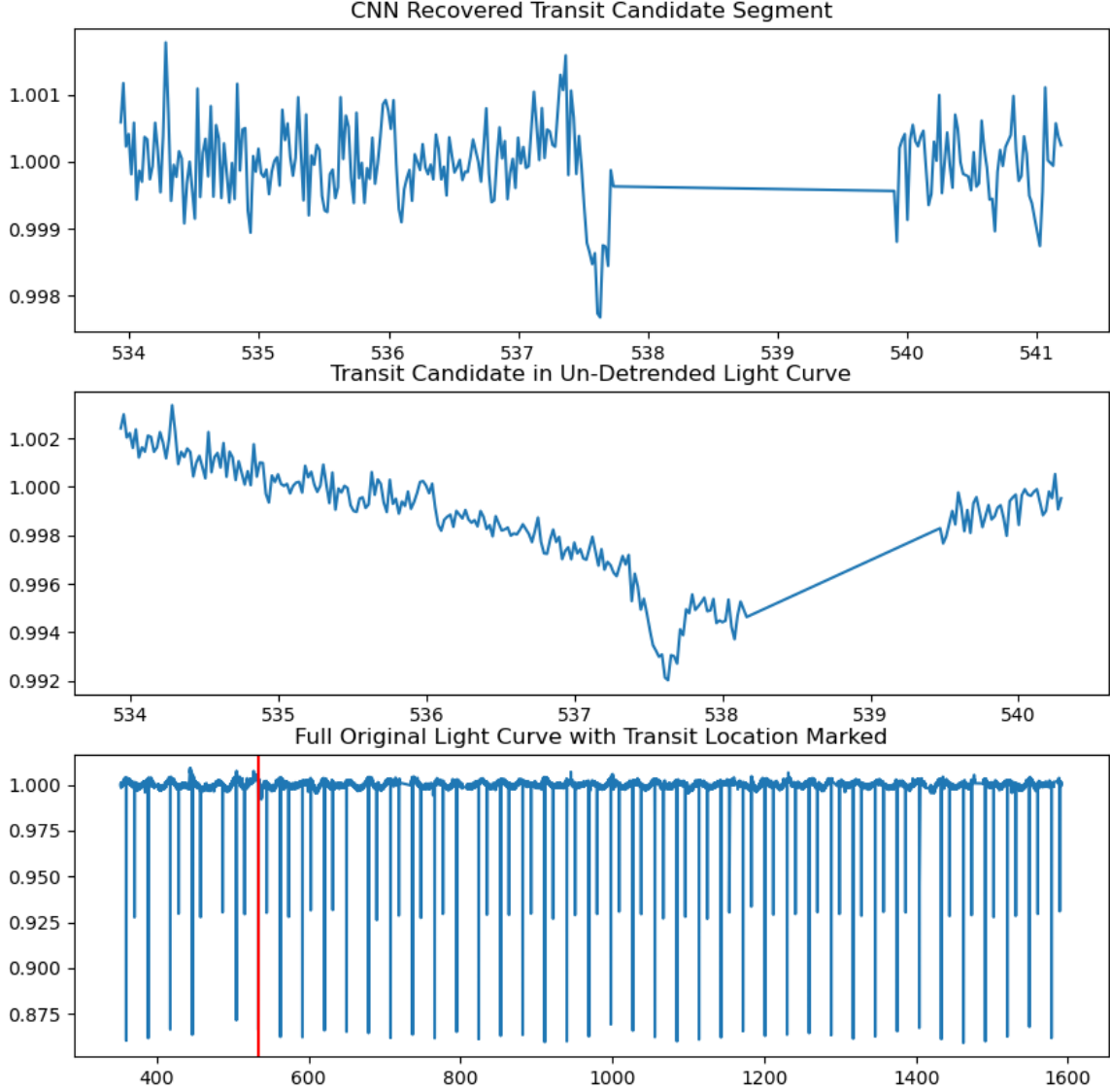


Figure 8. The light curve segment containing the identified transit candidate in KIC 7136958. Both the detrended light curve used to identify the candidate and the un-detrended original. The position of the candidate within the full original light curve is also marked.

low-SNR transit signals. Observation of rejected light curves indicates the CNN is mostly successful in excluding such areas in the blind search, but they are excessively common in the targeted 3σ selection dataset, which is due to many points in the regions exceeding global 3σ . These occurred frequently in both true and false positive systems.

An additional source are background binaries blended in with the light curve of the system of interest. When an additional eclipsing binary is located very close to a KIC system, the stellar flux measurement is a blend of the two (Bienias et al. 2021) and following the removal of the target’s stars the contaminating system’s features remain in the light curve. Eclipses in a contaminating system that is an eclipsing binary itself are readily detected by the model, since they are real transit features. These and the final type of false positive constitute the majority of my ‘high-confidence’ sample for this reason, but they can be distinguished by the complete lack of TTV and periods unphysical for circumbinary planets.

Lastly, a number of systems lack an associated KOI entry and associated masking information. Most of these are rotating ellipsoidal variables and overcontact systems which do not contain distinct eclipses and thus do not register in KOI automatic detections, others are semidetached systems or ‘heartbeat stars’ (pulsating tidal variables) (Thompson et al. 2012), while finally a few are ordinary eclipsing binaries which happen to lack a KOI entry. Most of the



Figure 9. The light curve segment containing the identified transit candidates in KIC 7668648. Both the detrended light curve used to identify the candidate and the un-detrended original. The position of the candidate within the full original light curve is also marked. Both of Transit Candidate 2’s outputs have been truncated by 2 days to avoid including a primary eclipse.

former type are eliminated by the preprocessing 5-day limit, but larger stars evidently can achieve a overcontact or semidetached configuration at longer periods. Outlier masking can usually remove eclipses even if there is no KOI record but can fail if eclipses are small or nonexistent compared to the ellipsoidal variations, as is sometimes the case in semidetached systems and nearly always the case in contact binaries.

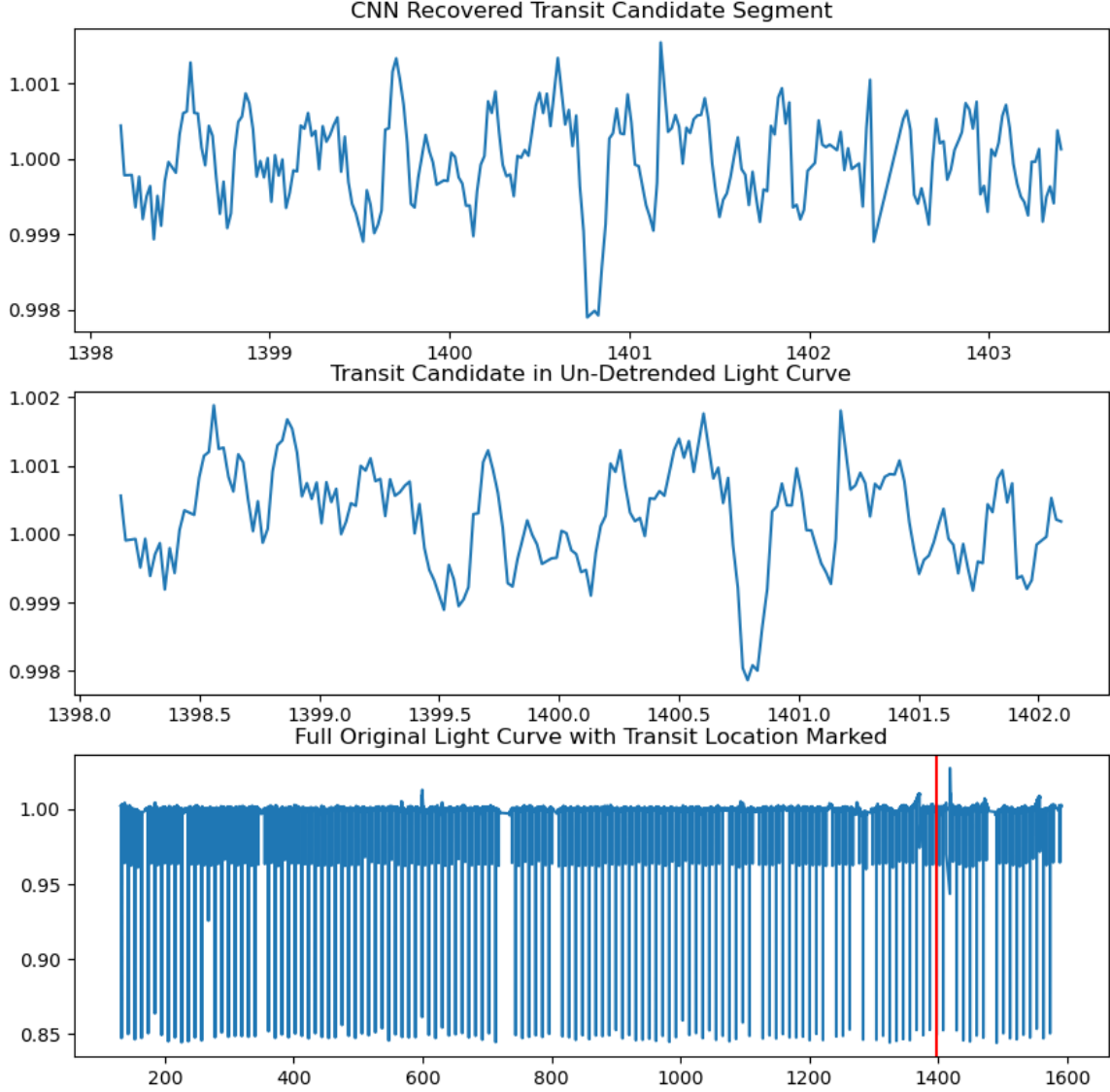


Figure 10. The light curve segment containing the identified transit candidates in KIC 11619964. Both the detrended light curve used to identify the candidate and the un-detrended original. The position of the candidate within the full original light curve is also marked. The un-detrended sample has been shifted right by 0.5 days to avoid including a masked secondary eclipse.

6. DISCUSSIONS

6.1. Comparative Method Performance

My survey is the largest survey of Kepler eclipsing binaries targeted specifically at the detection of circumbinary planets thus far and the only one to implement a large-scale CNN-based scanning approach. I was able to reduce the examination volume by 49.34% with respect to a visual-only survey, a significant improvement even in lieu of further reductions in false positives that could be achieved via retraining. This experimental approach suggests that even a rudimentary CNN is a feasible approach for detecting non-periodic or quasi-periodic transiting features like those produced by circumbinary planets.

The main issues encountered by the CNN in this experiment were unmasked targets, excessive noise, and blended background sources. Two of these sources of error were not the result of inaccurate classification, while the other could likely be resolved with additional training. Additional, CNN-assisted masking protocols may enable the elimination

Table 3. Candidate Object Radii

KIC Number	Candidate’s Estimated Radius (Jupiter Radii)
5983532	0.28745
6672229	0.73443
7136958	0.88513
7668648 Candidate 1	0.55063
7668648 Candidate 2	0.64295
7668648 Candidate 3	0.52997
11619964	0.65942

NOTE—Estimates of planetary size based on observed transit depth and stellar luminosity from Gaia.

of the former two sources of erroneous detections altogether, resulting in accuracy greatly improved compared to the present value.

6.2. Other Applications

The ability of this CNN to readily detect non-periodic events within noisy data suggests potential for the recovery of other non-periodic flux variations of interest to researchers. Investigations into stellar flares in exoplanet host systems have important implications for atmospheric erosion and hence habitability (Ridgway et al. 2023), and the non-periodic nature of stellar flares makes detection especially at low luminosities difficult. Applying a similar CNN-based approach might make such investigations more successful, allowing for improved modelling of planetary atmospheric erosion in prospective habitable systems. Non-repeating transits around single stars could be found using the same approach, identifying targets for follow-up observations and constraining the populations of long-period planets. Lastly, many stars are known to experience irregular dimming events for a variety of uncertain causes. Lower-magnitude analogues to the anomalous event at ‘Tabby’s Star’ or ‘Boyajian’s Star’, KIC 8462852 (Wright and Sigurdsson 2016) may be detected with CNN methods, while other dimming events such as exocomets (Kiefer et al. 2023) may also be recovered at high volumes with this method.

7. CONCLUSIONS & FURTHER RESEARCH

In summary, I have made the following achievements in my research using a purely synthetic dataset creation algorithm, convolutional neural network, and eclipsing binary processing pipeline:

1. I have trained a CNN with artificially synthesised data to recognize planetary transits within the light curves of eclipsing binaries. This CNN achieves relatively high levels of accuracy at very high speed, allowing for substantial reductions in manual search time. The pipeline I constructed with it is insensitive to problems such as TTV and transit size in comparison to stellar eclipses, allowing it to perform in situations where non-CNN automated of manual searches may fail.
2. I have performed a full CNN-driven search of 1143 eclipsing binaries in the Kepler primary mission survey field, out of a total of 2920 (39.14%), specifically targeted at those likely to host planets. This is the largest survey of Kepler eclipsing binaries for the purposes of circumbinary planet detection of any kind thus far.
3. I have recovered 5 tentative new circumbinary exoplanet candidates, a significant proportion of the present catalogue of known circumbinary exoplanets and the largest number of novel candidates identified in any single study. Though all cannot be confirmed with present data, the preliminary success of this method is nonetheless significant.

Having detected five novel circumbinary transiting planet candidates, I demonstrate the efficacy of my method and its preliminary applicability to the problem of detecting circumbinary planets in large volumes of Kepler flux data. Further refinement of the CNN and masking procedure is expected to significantly improve false positive exclusion in future versions of the model, but its present status is already sufficient to recover at least a few objects. While it is

not possible to constrain the orbital and physical parameters of the candidates recovered to any degree of certainty without additional observations, the detection of these singular events also demonstrates an applicability to completely non-periodic flux variations such as exocometary transits. It may also be applicable to the search for high-inclination circumbinary planets as are predicted by some studies (Cuello and Guippone 2019), but this applicability remains in principle and has not yet been tested.

Ultimately, CNNs represent an intriguing new possibility for expanding the search for circumbinary planets. As additional members of this exclusive class of worlds is discovered, our understanding of a particularly exotic mode of planetary formation will develop. Perhaps the contrasting nature of their populations compared to single-star systems may reveal critical insights into the planetary formation process, or habitable circumbinary worlds may be discovered. In either case, further investigation into these rare systems is surely to improve humanity’s understanding of the varied planetary architectures beyond our Solar System as a whole.

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